

CYCLISTIC DATA ANALYSIS PROJECT DOCUMENTATION

1. Ask Phase

Business Task:

Analyze Cyclistic's historical trip data to identify trends in member and casual rider behaviors. The goal is to understand how casual riders and annual members use Cyclistic bikes differently, and **to develop strategies to convert more casual riders into annual members.**

Key Stakeholders:

- ✚ Executive team
- ✚ Lily Moreno, Director of marketing
- ✚ Cyclistic marketing analytics team

2. Prepare Phase

Data Source:

Cyclistic's historical trip data (Divvy datasets) available through Motivate International Inc.

✚ **Link:** <https://divvy-tripdata.s3.amazonaws.com/index.html>

Data Type and Format:

CSV files containing trip details.

Data Range and Size:

Previous 12 months of data (selected quarters)

✚ **Selected Data:** {Divvy_Trips_2019_Q2; Divvy_Trips_2019_Q3; Divvy_Trips_2019_Q4; Divvy_Trips_2020_Q1}

Data Credibility:

The data is provided by a trusted public source (Divvy/Motivate), ensuring reliability and transparency. Also, no personal identifiable information included.

Licensing or Access Restrictions:

The dataset is made available under the terms outlined by Motivate International Inc.

Tools Used:

Microsoft Excel.

Initial Observations:

Upon examining the four quarterly datasets, the overall data reliability based on the **ROCCC framework** (Reliable, Original, Comprehensive, Current, and Cited) is acceptable. However, several structural and consistency issues were identified:

- ✚ **Schema Inconsistency:** The datasets have **non-uniform column headers** across quarters. For example, some files use **ride_id** while others use **trip_id** to represent the same variable.
- ✚ **Missing Attributes:** Certain quarters include additional variables such as **birthyear** and **gender**, while others do not. This creates a **variable inconsistency** that affects data merging and completeness.

- ✚ **Categorical Inconsistency:** The **user type variable** is labeled differently across datasets, some used **member/casual**, while others used **subscriber/customer**. This inconsistency will require **standardization** before analysis.
- ✚ **Structural Uniformity:** Despite these discrepancies, the datasets share a similar structure (ride-level trip data) and **will be integrated after cleaning and column harmonization**.

3. Process Phase

Data Cleaning Steps:

- ✚ **Removed Irrelevant Columns**

To maintain uniformity across all quarterly datasets, columns that were inconsistent or irrelevant to the analysis were removed. These include `rideable_type`, `birthyear`, `gender`, and other redundant variables.

- ✚ **Standardized Column Headers and Formats:**

Ensured consistent naming conventions and data types across all quarterly datasets to allow smooth merging and aggregation.

- ✚ **Standardized Categorical Variables:**

The `user_type` field was cleaned and standardized to contain only two consistent categories:

- Member (previously “subscriber”)
- Casual (previously “customer”)

This ensures accurate grouping and comparison during analysis.

Data Transformation:

- ✚ **Created a ‘ride_length’ Column:**

Calculated the duration of each trip by subtracting the start time from the end time:

- ✚ **Created a ‘day_of_week’ Column:**

Generated a numeric representation (1–7) for the day each ride started using the formula `=WEEKDAY(C2)` where: 1 = Sunday, 2 = Monday, ... 7 = Saturday.

This variable helps in identifying daily usage trends and behavioural patterns.

- ✚ **Converted Ride Duration to Numeric Format:**

To enable quantitative analysis, an additional column (`ride_length_minute`) was created by converting the time-based `ride_length` into numerical values (minutes).

This supports statistical operations such as averaging and comparisons.

- ✚ **Converted Month Data to Text Format:**

A new column (`month_text`) was derived from the `started_at` field, converting numeric month values (e.g., “1”) into readable text (e.g., “January”).

This enhances interpretability during visualization and reporting.

- ✚ **Combine quarterly datasets into one master dataset**

Data Validation and Consistency Checks:

- ✚ Verified that the ride_length values were positive and within reasonable limits (no negative or zero durations).
- ✚ Confirmed that all date and time fields used a consistent format across datasets.
- ✚ Ensured uniform data types across merged files to avoid analytical errors.

Tools:

All data processing activities were performed using Microsoft Excel.

Outcome:

The dataset was successfully cleaned, transformed, and standardized. It is now structured and ready for descriptive and exploratory analysis in the next phase

4. Analyze Phase

Exploratory Data Analysis (EDA)

The analysis focused on identifying behavioral and usage differences between annual members and casual riders using descriptive statistics and comparative analysis.

Key analytical activities included:

- ✚ Computing total rides count for annual members and casual riders to assess usage frequency.
- ✚ Calculating average ride length for both rider types to compare trip duration behavior.
- ✚ Analyzing average ride length by day of the week and user type to identify temporal usage patterns.
- ✚ Evaluating ride count distribution by day of the week to determine peak usage periods.
- ✚ Identifying frequently used start stations to understand spatial usage trends.

Visualizations

To support exploratory analysis and interpretation, the following Excel-based visualizations were developed:

- ✚ Average ride length of the week, segmented by rider type.
- ✚ Ride count by day of the week, segmented by rider type.
- ✚ Ride count by start station to highlight high-traffic locations.

Key Findings

- ✚ Annual members record a substantially higher total number of rides than casual riders.
- ✚ Casual riders have a higher average ride duration compared to annual members.
- ✚ Annual members' highest ride frequency occurs on Tuesdays, indicating dominant weekday usage.
- ✚ Casual riders' highest ride frequency occurs on Sundays, reflecting weekend-focused usage.
- ✚ Ride frequency and ride duration vary significantly by user type and day of the week.

Analytical Techniques

- ✚ Descriptive statistics, including averages, counts, and modes.

- ✚ Group-by comparisons across user type and day of week.
- ✚ Aggregation of ride-level data for station and temporal analysis.
- ✚ Interpretation of dashboard to identify behavioral trends.

Interpretation

The findings indicate that annual members primarily use Cyclistic bikes for routine, utilitarian purposes, such as commuting or regular weekday travel. This is supported by their higher ride frequency, shorter ride durations, and peak usage on a weekday, specifically Tuesday.

In contrast, casual riders demonstrate behavior consistent with recreational or discretionary use. Their longer average ride durations and peak usage on Sundays suggest leisure trips, sightseeing, or social activities concentrated on weekends.

These behavioral differences directly inform the business objective of increasing annual memberships by highlighting what casual riders value and how their needs differ from those of existing members.

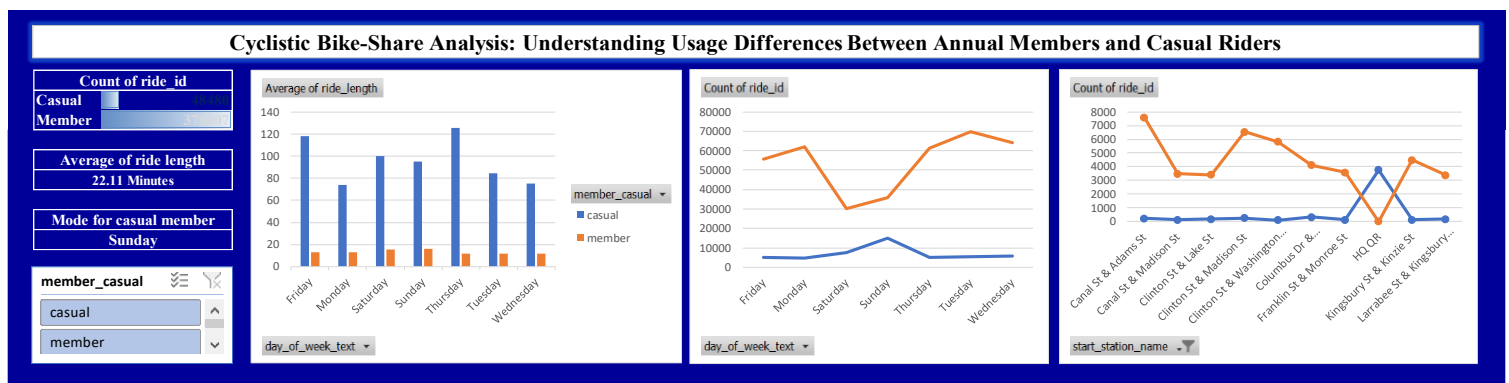
5. Share Phase

Dashboard Overview

The dashboard is designed to compare usage patterns between annual members and casual riders, highlighting key behavioral differences in ride frequency, duration, and timing. It is intended for Cyclistic's marketing and strategy teams to support data-driven decisions aimed at converting casual riders into annual members.

The dashboard informs decisions on targeted marketing campaigns, optimal engagement timing, and station-focused promotional strategies.

The Dashboard:



Key Dashboard Components

- ✚ Ride Count by User Type and Day of Week
- ✚ Average Ride Duration by User Type
- ✚ Usage Concentration by Start Station
- ✚ Temporal Usage Patterns Across the Week

Summary Insights

- ✚ Annual members primarily use Cyclistic bikes as a routine mobility solution, consistent with commuting or regular weekday travel.
- ✚ Casual riders are more likely to use bikes for leisure or discretionary activities concentrated on weekends.
- ✚ The longer ride durations of casual riders suggest higher perceived enjoyment or exploration rather than time-sensitive travel.
- ✚ Peak Sunday usage among casual riders represents a high-impact conversion window for targeted membership messaging.
- ✚ The strong weekday dependence of members highlights the value of predictable, repeat usage enabled by annual subscriptions.

6. Act Phase

Recommendations

Based on the analysis, the following data-driven marketing strategies are recommended:

✚ Weekend-to-Membership Conversion Campaigns

Since casual rider's peak on Sundays, Cyclistic can deploy targeted digital promotions during weekends. These could include in-app or email messages highlighting the cost savings of membership compared to repeated single-ride usage.

✚ Leisure-to-Lifestyle Messaging

Marketing content should reframe annual membership as not only a commuter benefit but also a lifestyle choice. Emphasizing unlimited access for weekend leisure rides can appeal to casual riders with longer trip durations.

✚ Short-Term Membership Trials

Offering discounted weekly or monthly trial memberships promoted on Sundays could reduce commitment barriers and encourage casual riders to experience member benefits firsthand.

✚ Station-Based Digital Promotions

High-traffic start stations used frequently by casual riders can be leveraged for geo-targeted digital ads or QR-based promotions that communicate membership value at the point of use.

✚ Value Comparison Visuals in Digital Media

Digital channels should clearly illustrate the difference in cost per ride between casual usage and annual membership, using simple, data-backed comparisons that resonate with frequent weekend riders.